

Total No. of Questions : 10]

SEAT No. :

[Total No. of Pages : 3

P3105

[5461]-142

B.E. (Mechanical)

POWER PLANT ENGINEERING

(2012 Course) (End Sem.) (402047) (Semester-II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Draw neat diagrams wherever necessary.*
- 3) *Use of scientific calculator is allowed.*
- 4) *Use of steam Table and Mollier Chart is allowed.*
- 5) *Assume suitable data wherever necessary.*
- 6) *Figures to the right indicate full marks.*

Q1) a) What is Carbon Credits? What is its significance in current energy scenario?[4]

b) A thermal power plant of 210 MW capacity has a maximum load of 160 MW. Its annual load factor is 0.6. The total coal consumption is 1 kg per kWh of energy generated and the cost of coal is Rs. 450.00 per tonne. Calculate: [6]

- i) The annual revenue earned if energy is sold at Rs. 1 per kWh and
- ii) The capacity factor of the plant.

OR

Q2) a) What are the element which contribute the cost of electricity? [4]

b) A power plant has the following annual factors: load factor = 0.75, capacity factor = 0.60, use factor 0.65. Maximum demand is 60 MW. Estimate: [6]

- i) The annual energy production,
- ii) The reserve capacity over and above the peak load, and
- iii) The hours during which the plant is not in service per year.

P.T.O.

- Q3) a)** What are the good requirements of coal handling plant? [4]
- b) Steam at a pressure of 15 bar and 250°C is expanded through a turbine at first to a pressure of 4 bar. It is then reheated at constant pressure to the initial temperature of 250°C and is finally expanded to 0.1 bar. Draw the T-S diagram and estimate the work done per kg of steam flowing through the turbine and amount of heat supplied during the process of reheat. Compare the work output when the expansion is direct from 15 bar to 0.1 bar without any reheat. Assume all expansion processes to be isentropic. [6]

OR

- Q4) a)** Enlist the advantages and disadvantages of Hydro Electric Power Plant (HEPP). [4]
- b) Draw a neat diagram of CANDU type of reactor and give its advantages and disadvantages over other types of reactors. [6]
- Q5) a)** Explain with neat sketch of diesel power plant showing all system used. [8]
- b) In a constant pressure open cycle gas turbine air enters at 1 bar and 20°C and leaves the compressor at 5 bar. Using the following data: temperature of gases entering the turbine = 680°C , pressure loss in the combustion chamber = 0.1 bar, compressor efficiency = 85%, turbine efficiency = 80%, combustion efficiency = 85%, $\gamma = 1.4$ and $C_p = 1.024 \text{ kJ/kgK}$ for air and gas, find: [8]
- i) The quantity of air circulation if the plant develops 1065 kW.
 - ii) Heat supplied per kg of air circulation,
 - iii) The thermal efficiency of the cycle.
- Mass of fuel may be neglected.

OR

- Q6) a)** Explain with details the Intercooling, reheating and regeneration of gas turbine power plant. [8]
- b) What are the advantages and disadvantage of Diesel power plant? [8]

Q7) a) Explain with neat sketch of solar pond and solar flat plat collector. [8]

b) Explain with neat sketch of Hybrid and Tidal power plant. [8]

OR

Q8) a) What are the different challenges in commercialization of nonconventional power plant? [8]

b) Write short note on: [8]

i) Fuel cell,

ii) Geothermal power plant

Q9) a) Explain different types of switch gears used in power plant electrical circuits. [9]

b) State any six effects of thermal pollution. Name any three source of thermal pollution. [9]

OR

Q10)a) Write short notes on: [9]

i) Circuit breaker

ii) Control System

iii) Causes of short circuits

b) Write short notes on: [9]

i) Global warming

ii) Noise pollution

iii) Green house effect

